

Consistency, Efficiency, and Scalability in Design Systems: A Comparative Analysis of Material Design, Apple's Human Interface Guidelines, and IBM Design Systems.

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This research paper aims to answer the question: 'How do Material Design, Apple's Human Interface Guidelines, and IBM Design Systems compare in terms of factors contributing to Consistency, Efficiency, and Scalability?' By employing a comparative analysis, this study offers valuable insights into the design systems. Conducting a thematic analysis of an autoethnographic journal, the findings provide a comprehensive look into the dynamics of Material Design, Apple's Human Interface Guidelines, and IBM Design Systems. A thematic analysis, paired with the perspective of Activity Theory, reveals a unanimous priority towards consistency across the systems. Yet, this often comes at the expense of potentially stifling designers' creative freedom. While each system boasts unique efficiency strategies, there's an overarching concern about generic solutions that might overlook specific user needs. Their adaptability to the fast-paced tech world underscores their scalability, but evolving designs may alienate users acquainted with older interfaces. Beyond these systems, the broader design community, encompassing designers to end-users, finds its practices and trends shaped by them. Delving deeper through an autoethnographic lens, the study offers a personal touch, spotlighting the profound societal ripple effects of design choices. Ethical quandaries emerge, pondering the balance between inclusivity and unintentional biases. In conclusion, while the research illuminates critical facets of these design systems, it also emphasises the fluidity of design perceptions and the importance of reflection in both design and research practices.

CCS CONCEPTS • Human-Computer Interaction

Additional Keywords and Phrases: Design systems analysis, Consistency, Efficiency, Scalability, Thematic Analysis

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1. INTRODUCTION

In today's digital landscape, design systems have emerged as a pivotal framework for companies to establish consistent user experiences across diverse platforms. Yew et al. [2020] highlighted that over 105 million web pages implemented at least one component from Google's Material Design. Within the HCI community, the allure of design systems isn't merely their ubiquity but also their profound impact on user experience. Effective design systems act as an essential bridge, ensuring users face no discontinuity when transitioning between interfaces, thereby removing the confusion that arises from inconsistencies or unfamiliar cues.

Consistency, efficiency, and scalability aren't just buzzwords in design; they are principles deeply rooted in academic understanding and industry practice. Lidwell, Holden, & Butler [2003] in their work "Universal Principles of Design", specifically argue that consistency is key to fostering usability and learnability. In contrast, efficiency, as discussed by Nielsen [1994] in "Usability Engineering," emphasises reducing user effort whilst maintaining a high level of productivity. Lastly, scalability in a design context can be understood as the ability of a system to handle growing amounts of work or its potential to be enlarged to accommodate that growth [2].

Despite their recognized significance, there remains a noticeable gap in academia: a thorough comparative evaluation of leading design systems, especially in relation to consistency, efficiency, and scalability. This research becomes essential because it offers guidance to designers in choosing the most appropriate system, specific to their needs and context.

The research's significance lies in its focus on critically examining design systems through an autoethnographic lens. This approach combines personal reflection and experience with academic understanding, offering a rich, multifaceted perspective. By positioning my personal experiences within broader HCI discourses, this study contributes a unique narrative that intertwines practice, challenges, and decision-making processes. While autoethnography has been employed in various fields for its ability to provide deep insights[9], its application in evaluating design systems offers fresh, academically sound perspectives that transcend traditional analytical boundaries. This study, through its unique methodological approach, moves beyond mere industry trends, aiming to make a substantial contribution to both academic discourse and practical design applications. Therefore, the research question driving this study is:

"How do Material Design, Apple's Human Interface Guidelines, and IBM Design Systems compare in terms of factors contributing to Consistency, Efficiency, and Scalability?"

By undertaking this research question, this study seeks to provide insights that are academically sound and practically relevant, assisting designers in crafting strategies that are both effective and align seamlessly with their design visions.

2. BACKGROUND AND THEORY

This section provides an overview of design systems and their advantages, along with explaining the terms of Consistency, Efficiency and Scalability, that are of utmost relevance for the research. Moving on to introducing Activity Theory as the theoretical framework of the study. It also introduces Material Design, Apple's Human Interface Guidelines and IBM Design Systems, the three design systems that are subject for this study.

2.1. Design Systems

As previously mentioned, design systems offer significant advantages by promoting consistency, efficiency, and scalability through the use of reusable components, which can help companies save time and resources [21, 22, 8]. The choice of Material Design, Apple's Human Interface Guidelines, and IBM Design Systems for this research is driven by their widespread adoption and influence in the design industry. Specifically, a survey conducted by Yew et al. at a design systems conference identified these as the top three most frequently used or tried design systems, excluding in-house built ones [24].

2.1.1 Design Systems and HCI

Design systems sit at the intersection of design and Human-Computer Interaction (HCI). HCI focuses on understanding users and designing technologies that enhance their experience, emphasising user-centricity, effectiveness, and adaptability[2]. The essence of design systems aligns with these principles. Furthermore, the idea of reusable solutions has been present in design and development through concepts like design patterns, which offer solutions to common design problems[10]. Building on this principle of reusability and standardisation, design systems evolved, providing more comprehensive guidelines and components that emphasise user-centricity, repeatability, and adaptability in digital design.

2.1.2 Key Aspects of Design Systems: Consistency, Efficiency, Scalability

The criteria of consistency, efficiency, and scalability were selected because they are pivotal to the evaluation of any design system. These metrics were identified through a review of literature as the most frequently cited parameters affecting the effectiveness of a design system [8, 18, 22]. The following explanation of the key terms, *consistency*, *efficiency*, and *scalability* in design systems is informed by Suarez [2019], Yew et al. [2020], and Edelberg & Kilrain [2020]. Drawing from these collective contributions, consistency is achieved by providing a set of standardised visual user interface elements that can be used across different platforms, leading to a cohesive and unified user experience that can improve brand recognition and user trust. In a design system, elements like buttons, typography, colour schemes, and icons are carefully defined and documented, ensuring a consistent look and feel throughout the product ecosystem. By adhering to these established design guidelines, companies can create a seamless user experience that fosters familiarity and enables users to navigate different

products or platforms effortlessly. This consistency not only enhances usability but also strengthens the brand identity, establishing a visual language that resonates with users.

Similarly, efficiency, as elaborated by the aforementioned authors, focuses on providing a library of pre-designed and reusable components, which designers can rapidly assemble user interfaces without starting from scratch for each project. This eliminates the need to recreate common elements and empowers designers to focus on solving unique design challenges. The time saved through component reuse allows for faster iteration and speeds up the overall product development process. Moreover, design systems promote collaboration and knowledge sharing by providing a centralised resource for designers to access and contribute to a shared repository of design assets and best practices. Scalability is fundamental to design systems' long-term viability. As noted by Suarez [2019], Yew et al. [2020], and Edelberg & Kilrain [2020], design systems can accommodate evolving needs by scaling seamlessly. They provide a flexible foundation that can adapt to changing requirements, ensuring design consistency and efficiency across a diverse range of projects. By incorporating modularity and flexibility into the design system's architecture, companies can effectively extend their design language to accommodate new platforms, applications, and functionalities.

2.2. Activity Theory in HCI

Activity Theory (AT) provides a robust framework for understanding how people interact with technology within the broader contexts of their lives. Key figures like Nardi [1996] and Kuutti [1996] have emphasised the significance of AT in HCI for its capacity to uncover intricate user behaviours and relationships with technological tools. The theoretical framework of this study is inspired by Activity Theory (AT). While AT traditionally encompasses a wide range of components including 'Subject,' 'Object,' 'Tools,' 'Rules,' 'Community,' and 'Division of Labor,' this research selectively employs the 'Subject,' 'Object,' 'Tools,' and 'Rules.' The focus on these particular components allows for a streamlined analysis that centres on the user (Subject), the design systems as mediators (Tools), the inherent guidelines and principles as 'Rules,' and the ultimate aim of achieving design consistency, efficiency, and scalability (Object). Components like 'Community' and 'Division of Labor' were excluded because they extend beyond the individual interaction with the design systems and delve into organisational and sociocultural dynamics, which are not the primary focus of this study.

The core components of AT include:

- Subject: The individual or collective embarking on an activity.
- Object: The ultimate goal or motive of the activity.
- Tools: Mediators the subject uses, such as design systems in our scenario.
- Rules: The set guidelines, both explicit and unspoken, which govern the activity.
- Community: The broader collective with intertwined objectives or goals.
- Division of Labor: How tasks and responsibilities are distributed among community members.

Kuutti [1996] elaborates on the utility of AT in design, stressing the imperative to comprehend not just isolated actions but the wider societal and cultural factors influencing them. Viewing design systems through AT's lens allows us to perceive more than just a tool. It uncovers the broader landscape, revealing the users, their motivations, the governing rules, and the communities they inhabit.

This holistic perspective of AT is complemented by the depth offered by autoethnography. Whereas autoethnography dives deep into personal experiences, AT situates these experiences within a larger context, creating a synergy for comprehensive understanding. However, it is worth noting, as Nardi [1996] points out, that while AT offers a sweeping perspective, it may occasionally overlook certain individual subtleties. This underscores the value of pairing AT with methodologies like autoethnography. In essence, by harnessing Activity Theory to scrutinise design systems, this research aspires to blend AT's contextual richness with the intimate insights of autoethnography, presenting a holistic view of design systems' roles in users' lives [16, 19].

2.3. Material Design

Using a tool like Google's open-source design system, "Material Design" is seen to be the most popular way of creating a design system. Yew et al. [2020] discovered that internal design systems were the second most common design systems, with many companies creating their own system to boost their brand identity and establish consistency in their design patterns and code libraries. However, it is interesting to note the fact that many of the companies continued to use Material Design's component library and principles as a benchmark for their own design systems, further indicating the system's popularity. The current version, Material 3, offers pre-designed components that may be adjusted for your unique preferences, allowing anyone to freely assemble and maintain their own design systems. They also feature a set of style guidelines for typography, colour, iconography, and motion design [12]. These guidelines offer suggestions and best practices for developing visually attractive and interactive designs that adhere to Material Design principles, enabling designers to make sure their design system is both user-friendly and aesthetic.

2.4. Apple's Human Interface Guidelines

Apple's Human Interface Guidelines (HIG) provide a comprehensive set of design principles and guidelines for creating user interfaces across Apple's ecosystem of devices, including iOS, macOS, watchOS, and tvOS. The HIG focuses on creating intuitive and engaging user experiences that align with Apple's design philosophy. The HIG covers various aspects of design, including visual design, interaction design, and user experience considerations. It offers guidance on typography, colour, iconography, layout, and navigation, ensuring consistency and coherence in the design of Apple's products. By following these guidelines, designers can create user interfaces that are familiar to Apple users and provide a seamless experience across different devices. The HIG also emphasises the importance of accessibility, providing recommendations for designing inclusive interfaces that accommodate diverse user needs. It highlights the use of assistive technologies, such as VoiceOver and Dynamic Type, to ensure that interfaces are usable by individuals with disabilities. Designers and developers can access the Apple HIG through the official Apple Developer website[1]. The guidelines are regularly updated to reflect the latest design trends and technological advancements, ensuring that designers stay aligned with Apple's design principles.

2.5. IBM Design Systems

IBM Design Systems is an open-source design system developed by IBM to foster design consistency and collaboration across its products and services. It provides a framework of visual and user interface components, design principles, and guidelines that help designers create cohesive and user-centred experiences. The IBM Design Language emphasises the principles of simplicity, inclusivity, and purposeful innovation. It focuses on creating designs that are accessible, usable, and emotionally engaging for a wide range of users. The language incorporates IBM's design thinking approach, which involves understanding user needs, prototyping, and iterative refinement. The design system consists of various components, such as grids, colours, typography, icons, and interactive patterns. These components are designed to be modular and reusable, allowing designers to create consistent and efficient user interfaces. The IBM Design Language also provides guidelines for designing accessible interfaces and ensuring compliance with industry standards and regulations. Designers and developers can explore the IBM Design Language through the official IBM Design website, which provides documentation, resources, and real-world examples to support the implementation of the design system in different projects [13].

3. METHODOLOGY

The methodology section presents a comparative analysis approach combined with autoethnography as the primary method for generating insights and data. Additionally, a thematic analysis approach was employed to identify related categories in the data. The chapter ends with a walkthrough on how the methods were applied in the study.

3.1. Comparative analysis

This study employed a comparative analysis, whilst integrating the method of autoethnography, to explore the factors that contributed to the consistency, efficiency and scalability of Material Design compared to Apple's Human Interface Guidelines and IBM Design Systems. This method was particularly chosen to identify variances and commonalities between different design systems. Comparative analysis is a research method that involves systematically comparing and contrasting different phenomena to identify similarities, differences, and underlying patterns [14]. In the context of design systems, a comparative analysis helps uncover the unique characteristics of each system, by comparing how each system contributes to these three factors. The analysis was then conducted by examining and comparing these key aspects of the design systems. Firstly, the design principles of each system were evaluated to understand their alignment with the goals of consistency, efficiency, and scalability. This assessment considered personal experiences and insights gained through autoethnography, providing a unique perspective on the design systems. While autoethnography captures the researcher's personal interactions and reflections with the design systems, the comparative analysis serves to systematically contrast these experiences across different design systems. It is here in the comparative analysis that we attempt to extract more general insights from the deeply personal data generated through autoethnography.

3.2. Autoethnography

Autoethnography, also called personal ethnography, is an ethnographic method where the interest of the research lies in the personal experiences of the researcher. Autoethnography, at its core, is a subjective research method that relies heavily on the personal experiences of the researcher. These experiences, though deeply personal, serve as a lens through which broader phenomena can be understood. This method is especially beneficial for capturing nuanced insights based on my interactions with the design systems, thereby enriching the comparative analysis. In autoethnography, the primary aim is to glean insights about potential features and functionality through journaling and note-taking, capturing the researcher's personal behaviour during the activity of interest [7]. In contrast to traditional research methods, where the researcher is typically an observer of the research subject rather than the subject themselves, this approach places the researcher in a dual role as both the researcher and the research subject. The method was created as a response to complaints that personal experience is less regarded in analyses of more significant social and cultural phenomena in academic writing [4].

This observational aspect of autoethnography clearly aligns with the goal of generating insights into the design systems under investigation. By consciously observing and noting the personal encounters and interactions with Material Design, Apple's Human Interface Guidelines, and IBM Design Systems, the aim of the paper is to uncover patterns and themes that may not be evident through other research methods[17]. The observational nature of autoethnography allows for a deeper exploration of the researcher's subjective experiences, providing rich data that can contribute to a more comprehensive analysis and understanding of the design systems.

The autoethnographic data collection process involves journaling and note-taking throughout the duration of the study. The researcher will write down their thoughts, observations, and reflections during their engagement with the design systems. This method allows for a reflexive and iterative process, where the researcher critically reflects on their experiences, beliefs, and values, and examines how these factors influence their design choices and outcomes. By marrying the subjective insights from autoethnography with the structured approach of comparative analysis, this study offers a unique blend of personal narratives with broader design system evaluations. This fusion allows for a rich, multi-dimensional exploration of Material Design, Apple's Human Interface Guidelines, and IBM Design Systems.

3.3. Thematic Analysis

Thematic analysis is a recommended method to use to analyse the autoethnographic journal. By generating themes and relevant categories, we create meaning and explanation of the data[5]. A thematic analysis can be done in one of two primary ways, either inductively (bottom up) or deductively (top down). With the use of the inductive method, the themes can develop outside of the researcher's theoretical interests or existing codes in a given field or topic. According to Braun and Clarke [2006], the themes observed through the inductive approach

should be closely related to the data and are coded based on the data themselves, rather than being driven by a pre-existing coding frame or the researcher's analytic preconceptions.

In contrast, the deductive approach is driven by pre-existing coding frames or theoretical interests. In this approach, the researcher starts with a set of predetermined themes or codes derived from existing theories or prior research[3] In this study, the deductive approach is particularly relevant as it aligns with the research question at hand: "What are the factors contributing to the consistency, efficiency, and scalability of Material Design, Apple's Human Interface Guidelines, and IBM Design Systems?" The themes of consistency, efficiency, and scalability, central to this research question, have been identified based on established research and prior knowledge in the field of design systems. These themes serve as a framework for analysing and interpreting the factors associated with the three design systems under investigation. The deductive approach enables a structured and systematic examination of the identified themes. Given the rich narrative data produced through autoethnography, thematic analysis was employed to lend structure and coherence to the results, making them easier to interpret. The researcher reviews the autoethnographic journal entries, coding the data based on the predetermined themes of consistency, efficiency, and scalability. This process ensures that the analysis remains closely tied to the data itself and is guided by the researcher's theoretical interests in understanding the factors contributing to the performance of the design systems.

3.4. Method application

The integration of these methods is designed to create a comprehensive analytical framework that merges empirical evaluation with personal insights. Comparative analysis provides the breadth, autoethnography adds depth, and thematic analysis offers the structure for data interpretation. To achieve the research objectives, the autoethnographic process was employed, enabling an immersive and reflective exploration of each design system. This was done by actively engaging with the Material Design, Apple's Human Interface Guidelines and IBM Design Systems respective web-resources, delving deep and looking into their principles for each of the three core themes. This immersion allowed for a firsthand understanding of the design systems' characteristics. Throughout this engagement with each design system, I maintained a detailed journal, documenting personal observations and reflections. It provided means to capture my initial impressions and moments of insight for each design system. The journal entries were written in a narrative style, allowing me to express my thoughts and emotions in a free-flowing manner. As the researcher conducting this autoethnographic study, I recognize the influence of my own biases, preconceptions, and subjectivity on the research process and findings. This is an aspect that the method has received criticism for, meaning that it is very important to acknowledge that my background, experiences, and personal perspectives may shape the interpretations and reflections presented in this study. By being aware of these factors, I strive to maintain a critical and reflexive stance throughout the research process. This transparency contributes to the overall rigour and credibility of the study [11].

I approached each design system with an open mind, engaging with Material Design, Apple's Human Interface Guidelines and IBM Design Systems. The journal became a repository for documenting my journey and served as a valuable tool for reflection. For consistency, I paid close attention to the visual cues, typographic styles, and colour palettes employed by each design system. As I explored the design systems, I made note of the recurring themes, patterns, and elements that contributed to a sense of consistency. These observations were documented in my journal, accompanied by personal reflections on the impact of consistency on user experience and brand perception. Next, I turned my attention to efficiency, delving into the pre-designed components, design thinking methodologies, and available resources offered by each design system. I highlighted specific instances where the use of reusable components or adherence to design thinking principles saved time and streamlined workflows. Finally, I explored the concept of scalability within each design system. I examined how the guidelines and components were designed to adapt seamlessly across different screen sizes, resolutions, and interaction patterns.

With each principle examined individually, my journal entries were filled with key words, thoughts, and insights that emerged during the process. Drawing from the wealth of information and reflections captured in my journal, I summarized my findings in a free-flowing text. This summary represented my experiences, thoughts, and

observations, providing a cohesive narrative that highlighted the unique characteristics and strengths of each design system in relation to consistency, efficiency, and scalability. The data generated through the autoethnographic process will undergo thematic analysis as previously described. I reviewed the journal entries, identified significant categories and patterns, and coded them based on the data. The coding process followed an inductive approach, where descriptive keywords were transformed into categories. The resulting categories were noted and presented in tables for analysis and discussion, facilitating reflection on the emerging categories within each theme. This coding method is referred to as micro-coding and refers to the process of analysing and categorising small units of data within a larger dataset, assigning specific codes or labels to these units, allowing for detailed analysis and identification of patterns.[5] The table below showcases how the data was turned into the categories.

Data from the autoethnographic journal	Coded Categories
<i>[For example, the thoughtful use of consistent visual cues like recognizable icons and button styles empowers users to navigate interfaces effortlessly, creating a strong sense of familiarity and ease of use]</i>	Recognizable icons and button styles
<i>[Apple achieves this through the use of responsive design techniques, enabling interfaces to fluidly adjust to various devices, from iPhones to iPads and Macs. For example, the HIG provides guidelines for creating adaptive layouts that automatically adjust content and maintain legibility across different screen dimensions]</i>	Adaptive Layouts
<i>[Furthermore, IBM Design Systems provide a range of design resources, such as design libraries, UI kits, and code snippets. These resources serve as valuable references for designers, offering pre-built components, design patterns, and coding guidelines]</i>	Reusable elements

Table 1: Micro-coding example

The combination of objective analysis and subjective experiences will provide a comprehensive and nuanced exploration of the design systems under investigation, shedding light on the factors contributing to their consistency, efficiency and scalability.

4. RESULTS AND ANALYSIS

The autoethnographic journal serves as the primary data source for this study. Data was generated throughout the work and the diary entries capture a wealth of thoughts, ideas and insights that emerged during the observations and interactions. Thematic analysis and Activity Theory is then applied to the collected data, uncovering any significant patterns and noteworthy findings. The chapter concludes with a summary.

4.1. Autoethnographic Results

This section presents the autoethnographic results based on my interaction and exploration of Material Design, IBM Design Systems and Apple's Human Interface Guidelines.

4.1.1 Material Design

Material Design has truly captured my attention with its remarkable characteristics. Consistency, which lies at the heart of this design system, is a key driver of its effectiveness. The guidelines of Material Design establish a standardised set of rules and principles that can be applied seamlessly across different devices and platforms.

This consistency greatly enhances the user experience, enabling individuals to quickly understand and interact with Material Design interfaces based on their prior exposure. For example, the thoughtful use of consistent visual cues like recognizable icons and button styles empowers users to navigate interfaces effortlessly, creating a strong sense of familiarity and ease of use. Moreover, Material Design employs a cohesive colour palette and typography, ensuring a harmonious visual experience throughout the site. The consistent application of typography styles across headings, paragraphs, and other text elements maintains a unified look and feel. This attention to consistency not only contributes to a visually appealing interface but also once again, establishes a sense of trust and familiarity. When users encounter consistent design elements, it instils confidence in the reliability and professionalism of the product or service. By emphasising consistency, Material Design sets a strong foundation for creating user-centric interfaces. It promotes a standardised language that enables designers to effectively communicate with users and facilitates intuitive interactions.

Efficiency stands out as a crucial aspect contributing to the widespread popularity of Material Design. The utilisation of pre-designed components, such as cards and lists, enables designers to expedite the prototyping and iteration processes, thereby streamlining their workflow. These reusable elements not only serve as a valuable starting point but also save significant time, allowing designers to allocate more resources to refine the overall design concept and enhance the user experience. This practical approach empowers designers to work more productively, delivering interfaces of high quality, even within demanding project timelines. Additionally, the Material Design website itself exemplifies efficiency through its thoughtful organisation and easily accessible information. With a clear navigation structure, intuitive search functionality, and concise descriptions of design principles and components, users can efficiently explore and grasp the core concepts of Material Design. The emphasis on providing concise yet comprehensive information enables designers to quickly familiarise themselves with the guidelines and effectively apply them in their projects, further enhancing their efficiency and effectiveness.

Scalability represents another remarkable strength of the Material Design system. The guidelines and components are purposefully designed to seamlessly adapt to various screen sizes, resolutions, and interaction patterns. The philosophy of responsive design ensures that the content and layout fluidly adjust to different device dimensions, providing an optimal viewing experience regardless of whether users access the interface on smartphones, tablets, or desktop computers. This scalability proves particularly crucial in the contemporary multi-device landscape, where users anticipate consistent and accessible experiences across diverse platforms. By leveraging Material Design's inherent scalability, designers can cater to the specific needs and constraints of their projects while maintaining a visually cohesive and interactive design language. It empowers them to create interfaces that seamlessly transition between different devices, offering users a unified experience without compromising functionality or aesthetics. This scalability not only future-proofs designs but also provides the flexibility required to accommodate evolving technological advancements and changing user behaviours.

In conclusion, Material Design's consistency, efficiency, and scalability are key factors driving its higher adoption and usage compared to other design systems. Its guidelines and components ensure a standardised and cohesive experience across devices and platforms, while also enabling designers to work more efficiently. By leveraging the pre-designed components and adhering to the guidelines, designers can create interfaces that strike a balance between adhering to a standardised design system and expressing their unique creativity.

4.1.2 Apple's Human Interface Guidelines

During my exploration of Apple's Human Interface Guidelines (HIG), I was once again struck by the strong emphasis on consistency as a foundational principle. Apple achieves this through meticulous attention to detail, ensuring that their interfaces maintain a cohesive visual language across their ecosystem. For instance, the HIG provides specific guidelines for typography selection, recommending the use of San Francisco as the primary typeface. This consistent typography creates a sense of familiarity and aligns with Apple's brand identity, allowing users to recognize Apple products instantly. Furthermore, the HIG emphasises the importance of consistent colour schemes and iconography, which not only enhance visual appeal but also facilitate seamless

navigation and interaction. By adhering to these guidelines, designers can create interfaces that consistently convey Apple's distinctive style and evoke a sense of trust and reliability.

Efficiency is another aspect that stands out in Apple's HIG. The guidelines provide practical recommendations to streamline interactions and optimise user experiences. Apple places great importance on using clear and concise language in interface elements, enabling users to quickly understand and accomplish tasks. For example, the HIG advises using descriptive button labels that convey the expected action, reducing ambiguity and minimising cognitive load. Additionally, Apple promotes the use of intuitive gestures, such as swiping and pinching, which align with users' natural behaviours. These intuitive interactions reduce the learning curve, allowing users to navigate and interact with interfaces more efficiently. By providing these well-defined interaction patterns, Apple empowers designers to create interfaces that prioritise task completion and enhance overall user productivity.

Scalability is a crucial consideration in today's multi-device landscape, and Apple's HIG addresses this aspect effectively. The guidelines provide comprehensive strategies for designing interfaces that seamlessly adapt to different screen sizes and resolutions. Apple achieves this through the use of responsive design techniques, enabling interfaces to fluidly adjust to various devices, from iPhones to iPads and Macs, even their Apple Watches and TVs. For example, the HIG provides guidelines for creating adaptive layouts that automatically adjust content and maintain legibility across all the different screen dimensions and products. Additionally, Apple emphasises the use of scalable vector graphics (SVGs) and flexible design components, allowing visual elements to scale without compromising quality or usability. This scalability ensures that interfaces remain visually appealing and functional across a range of devices, providing users with a consistent and accessible experience.

In conclusion, Apple's Human Interface Guidelines embody a design philosophy centred around consistency, efficiency, and scalability. The meticulous attention to detail regarding typography, colour schemes, and iconography ensures a cohesive visual experience that aligns with Apple's brand identity. The emphasis on clear and concise language, as well as intuitive interactions, optimises user efficiency and enhances task completion. Moreover, the comprehensive strategies for scalability enable interfaces to adapt seamlessly to diverse devices, ensuring consistent usability and visual appeal. By following Apple's HIG, designers can create interfaces that embody Apple's distinctive style while delivering exceptional user experiences.

4.1.3 IBM Design Systems

IBM Design Systems encompasses various components and resources that contribute to its efficiency. One prominent example is the Carbon Design System, a comprehensive design system with guidelines and components designed to streamline the creation of user interfaces across IBM's digital platforms. The Carbon Design System provides designers with reusable elements like buttons, forms, and data visualisations, enabling them to quickly assemble interfaces and reduce development time. This modular approach significantly enhances efficiency by eliminating the need to build every interface element from scratch, allowing designers to focus on higher-level aspects of the user experience. In addition to the Carbon Design System, another noteworthy aspect of IBM Design Language is the IBM Plex font. Developed specifically for IBM, the IBM Plex font family offers a unified and versatile typographic system that enhances visual consistency across digital and print materials. By employing a consistent and visually appealing font, IBM Design Language ensures that text elements are legible, harmonious, and aligned with IBM's brand identity. This consistency not only contributes to a cohesive design language but also enhances the overall user experience by providing familiarity and clarity.

Moreover, IBM Design Language promotes the use of design thinking methodologies, which emphasise user research, iterative prototyping, and user testing. By incorporating these practices into the design process, IBM Design Language enables designers to gain valuable insights into user needs and preferences. This user-centric approach fosters efficiency by ensuring that design decisions are based on empirical data rather than assumptions. Through iterative prototyping and user testing, designers can quickly identify and address usability

issues, leading to more efficient and effective design solutions. Furthermore, IBM Design Language provides a range of design resources, such as design libraries, UI kits, and code snippets. These resources serve as valuable references for designers, offering pre-built components, design patterns, and coding guidelines. By leveraging these resources, designers can save time and effort, accelerating the design and development process. This emphasis on providing comprehensive design resources reinforces the efficiency of IBM Design Language by equipping designers with the tools they need to create high-quality and consistent user interfaces.

To facilitate scalability, IBM Design Language provides guidelines and components that are purposefully designed to be flexible and adaptable. For example, the Carbon Design System offers responsive grid layouts, allowing designers to easily create interfaces that fluidly adjust to different screen sizes. Additionally, IBM Design Language promotes the use of scalable vector graphics (SVGs) and iconography, ensuring that visual elements can scale without loss of quality or legibility. By incorporating these scalable components and practices, IBM Design Language equips designers with the tools to create interfaces that can effectively accommodate various screen sizes and resolutions.

In conclusion, IBM Design Language incorporates various elements that contribute to its efficiency. The Carbon Design System, the IBM Plex font, design thinking methodologies, and a range of design resources all play key roles in streamlining the design process. By leveraging these components and practices, IBM Design Language empowers designers to work efficiently, delivering consistent and user-centric design solutions. These examples highlight the unique characteristics and how it addresses the challenges of consistency, efficiency, and scalability. By adopting IBM Design Language, designers can create interfaces that not only meet the needs of users but also align with IBM's brand identity and values.

4.2. Comparative Thematic Analysis

This section presents a thematic analysis based on my autoethnographic results from the interaction and exploration of Material Design, IBM Design Systems, and Apple's Human Interface Guidelines. The approach incorporates a balance between my personal experiences and the broader implications of design choices.

	Consistency	Efficiency	Scalability
Material Design	Recognizable icons and button styles, Cohesive colour palette and typography, Unified look and feel	Utilisation of pre-designed components, Thoughtful organisation, Easily accessible information	Responsive design, Adaptability to various screen sizes and interaction patterns, Future-proof design
Apple's Human Interface Guidelines	Typography, Color Schemes, Iconography	Clear and Concise Language, Intuitive interactions	Responsive Design, Adaptive Layouts, Scalable Vector Graphics
IBM Design Systems	Carbon Design System, IBM Plex font	Design thinking methodologies, Reusable elements, Design resources	Responsive grid layouts, Scalable vector graphics, Adaptable design components

Table 2: Themes and Categories for the design systems

In any design ecosystem, consistency, efficiency, and scalability serve as guiding principles that shape user experience. While Material Design, Apple's Human Interface Guidelines (HIG), and IBM Design Systems all pay homage to these principles, the way each system navigates them offers revealing insights.

The thematic analysis reveals a shared concern for consistency across all three design systems. Material Design opts for recognizable icons and a cohesive colour palette, aiming to reduce cognitive load. This suggests a user-centric philosophy geared towards immediate understanding, a likely response to the impatient nature of today's digital users. Apple's HIG, with its stringent adherence to San Francisco as the primary typeface, seeks

to provide intuitive engagement. This aligns with Apple's brand ethos of creating a closed, cohesive ecosystem where each element complements the other. This consistency allows for a more immersive experience, particularly crucial for keeping users within the Apple ecosystem. Meanwhile, IBM Design Systems employs its Carbon Design System and IBM Plex font to convey reliability. This seems strategically geared towards the enterprise sector where IBM mostly operates, underlining the firm's awareness of its primary audience's needs for dependable interactions.

While Material Design and Apple's HIG prioritise user intuition and efficiency, IBM's focus leans more toward brand consistency. The interesting part here is that each system is optimising for what they perceive as their core user's primary needs. This divergence could be attributed to the primary goal of each system: Material Design focuses on user familiarity, Apple on seamless and productive interaction, and IBM on reliability. This suggests a keen understanding of target demographics, each tailoring their approach based on user personas most prevalent in their respective ecosystems.

Efficiency also has varying ramifications across these systems. Material Design's pre-designed components can sometimes sacrifice depth for the sake of speed, offering more generic solutions. This raises the question of whether fast-to-market solutions are prioritised over highly customised experiences, an important consideration given the fast-paced world of app development. Apple's HIG, on the other hand, balances efficiency with a focus on user delight. This balance may be indicative of Apple's confidence in its consumer loyalty, allowing for more time-intensive design choices. IBM's design thinking methodologies, while efficient, could risk leading to designs that lack uniqueness. This efficiency comes at a cost; it may lead to enterprise solutions that are robust but not particularly innovative, a trade-off that IBM seems willing to make.

Scalability is another shared theme but with different emphases. Material Design and Apple's HIG give more weight to adaptability, preparing for various screen sizes and interaction patterns. This adaptability could serve as a crucial competitive advantage in emerging markets where device variety is more prevalent. IBM, although it incorporates scalability, does so with a more cautious approach. This could be interpreted as IBM's method to safeguard its brand uniformity, a highly prized attribute in its enterprise clientele.

Overall, the comparative thematic analysis illustrates that while all three systems share key thematic concerns, the manner in which these concerns are prioritised and executed differs significantly. This speaks to the larger strategic vision behind each design system, revealing not just design choices but also market positioning and business strategy.

4.3. Comparative Analysis with Activity Theory-Inspired Framework

Using an Activity Theory-inspired framework enriches our comparative of Material Design, Apple's Human Interface Guidelines (HIG), and IBM Design Systems by situating them within broader contexts.

Object (Goal of the Design Systems):

Material Design aims to promote seamless user interactions through recognizable visual cues, directly targeting the reduction of cognitive load. In contrast, Apple's HIG prioritises intuitive user engagement, streamlining interactions to focus on aspects such as user delight and productivity. This suggests that Material Design leans toward a utilitarian objective, while Apple aspires for a more experiential outcome. IBM Design Systems, meanwhile, eschew both approaches to place reliability and trustworthiness at the forefront, thereby establishing a unified design experience across products. This indicates a more risk-averse and enterprise-focused objective.

Subject (Users):

All three systems serve a dual audience of designers and end-users. However, the intimate insights from autoethnography reveal disparities. Material Design's broad palette and iconic elements may resonate with users seeking quick, intuitive interactions, but can risk alienating those who find its interface overly simplistic. Apple's HIG, with its focus on aesthetic and functional coherence, may delight many but perplex those unused to the Apple ecosystem. IBM's system, designed with reliability in mind, is most likely to appeal to enterprise users who prioritise that over design flair.

Tools (Design Elements):

Material Design primarily uses pre-designed components and navigational structures, offering a toolkit that accelerates design implementation but may lack depth. Apple's HIG employs typography rigorously (San Francisco font), along with concise language and intuitive gestures, to cultivate a distinct user experience. This suggests that Apple places a high premium on the tools used to convey its design philosophy. IBM Design Systems, on the other hand, lean heavily on the IBM Plex font and design thinking methodologies, emphasising reusable elements. This reflects a focus on standardisation and long-term sustainability in design elements.

Rules (Guidelines):

Each system has its unique set of guidelines, but there are critical divergences here too. Material Design's guidelines prioritise quick understanding and ease of use, sometimes at the expense of depth or creative latitude. Apple's rules aim to strike a balance between consistency and creative expressiveness, but this may demand more from designers in terms of time and cognitive load. IBM's guidelines, often grounded in extensive research, can be stringent, reflecting the brand's emphasis on reliability but potentially stifling creativity.

By adopting an Activity Theory-inspired approach, this analysis lends an extra layer of depth and context. It offers a structured way to understand how these design systems are not just tools but complex mediators that influence both designers and users, enabling a nuanced look into how these different systems serve or challenge their users, as shaped by each system's unique goals, user needs, design elements, and guidelines.

4.4. Comparative Summary and Implications

By integrating findings from the thematic analysis with the perspective of Activity Theory, several key insights and implications become apparent. All three design systems—Material Design, Apple's Human Interface Guidelines, and IBM Design Systems—prioritise consistency. However, a potential drawback is that the strict adherence to design guidelines might limit the creative flexibility of designers.

In terms of efficiency, each system has its approach, but there is a concern. The design elements provided can lead to generic solutions, which might not always cater to specific user needs. Regarding scalability, the systems demonstrate adaptability in the context of a rapidly changing technological environment. But there's a risk: as designs evolve, some users familiar with older interfaces might feel left out. At a broader level, the 'community'—which includes designers, developers, stakeholders, and end-users—has a critical role. Their practices and expectations are influenced by these design systems, impacting not just individual experiences but also setting trends in design.

To address the research question, "How do Material Design, Apple's Human Interface Guidelines, and IBM Design Systems compare in terms of factors contributing to Consistency, Efficiency, and Scalability?", it is clear that each system has its advantages and challenges. Striking a balance between maintaining a familiar user experience and introducing innovative designs, meeting diverse user needs efficiently, and updating designs without excluding some users are all challenges faced by these systems. With insights from both thematic analysis and Activity Theory, we can view these design systems as not just tools but key players in the broader design and user community.

5. DISCUSSION

Using an autoethnographic approach, this study delved into a personal exploration of the three design systems. By combining my experiences with analytical thoughts, I provided a unique viewpoint on these systems. However, this method has its limitations. Even though autoethnography offers a deep understanding, its findings are primarily based on my individual experiences. This means they may not capture the broader perspectives of the entire design community. While the revised analysis introduced an Activity Theory-inspired framework to add further context, the findings are still drawn from a single perspective, underscoring the need for more diverse user studies in future research.

In analysing my interactions with these systems, I recognized the important link between design choices and their wider social implications. These design systems, while guiding designers, greatly influence user behaviours and interactions. This raises ethical questions about their role. These systems can promote inclusivity and fairness, but they can also unintentionally introduce biases. Given the profound impact of design decisions on society, it is vital to be aware of these potential biases and work to minimise them.

Reflecting on this research, it is evident that my insights provide just one perspective on the design systems. Since design is a constantly evolving field, the way individuals perceive and use these systems might change over time. The popularity of a design system today doesn't guarantee its continued dominance tomorrow. However, this study offers clear insights for both designers and researchers. By applying an Activity Theory-inspired approach in the revised analysis, we've gained a multifaceted understanding, but the limited breadth of perspective should be taken into account when considering the implications. For designers, it highlights the need for careful thought and reflection when choosing and using design systems. For researchers, especially those using autoethnographic methods, it demonstrates the value of personal experiences in research, but also the importance of acknowledging the limitations of such a personal approach.

REFERENCES

- [1] Apple. n.d. Human Interface Guidelines. Retrieved from <https://developer.apple.com/design/>
- [2] A. B. Bondi. 2000. Characteristics of scalability and their impact on performance. In Proceedings of the 2nd international workshop on Software and performance (pp. 195-203).
- [3] V. Braun and V. Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77-101.
- [4] K. Butz and K. Besio. 2009. Autoethnography. *Geography Compass* 3, 5 (2009), 1660–1674. <https://doi.org/10.1111/j.1749-8198.2009.00279.x>
- [5] H. Chang, F. W. Ngunjiri, and K. A. Hernandez. 2012. Collaborative Autoethnography. Taylor & Francis Group. Retrieved from <http://ebookcentral.proquest.com/lib/uu/detail.action?docID=1110067>.
- [6] E. F. Churchill. 2019. Scaling UX with design systems. *Interactions* 26, 5 (Sept.-Oct. 2019), 22–23. <https://doi-org.ezproxy.its.uu.se/10.1145/3352681>
- [7] S. J. Cunningham and M. Jones. 2005. Autoethnography: a tool for practice and education. In Proceedings of the 6th ACM SIGCHI New Zealand chapter's international conference on Computer-human interaction: making CHI natural (CHINZ '05). ACM, New York, NY, USA, 1–8. <https://doi-org.ezproxy.its.uu.se/10.1145/1073943.1073944>
- [8] J. Edelberg and J. Kilrain. 2020. Design Systems: Consistency, Efficiency & Collaboration in Creating Digital Products. In Proceedings of the 38th ACM International Conference on Design of Communication (SIGDOC '20). ACM, New York, NY, USA, Article 8, 1–3. <https://doi-org.ezproxy.its.uu.se/10.1145/3380851.3416743>
- [9] C. Ellis, T. E. Adams, and A. P. Bochner. 2011. Autoethnography: An Overview. *Forum: Qualitative Social Research*, 12(1).
- [10] Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley. ISBN: 0201633612.
- [11] C. Ghita and C. Thorén. 2021. Going cold turkey!: An autoethnographic exploration of digital disengagement. *Nordicom Review* 42, 4 (2021), 152-167.
- [12] Google. 2022. Material Design. Retrieved February 14, 2023, from <https://material.io/design/>
- [13] IBM. n.d. IBM Design Language. Retrieved from <https://www.ibm.com/design/language/>
- [14] J. Li, Y. Li, H. Han, and X. Lu. 2022. Exploratory Methods for Imbalanced Data Classification in Online Recruitment Fraud Detection: A Comparative Analysis. In 2021 4th International Conference on Computing and Big Data (ICCBD 2021). ACM, New York, NY, USA, 75–81. <https://doi-org.ezproxy.its.uu.se/10.1145/3507524.3507537>
- [15] W. Lidwell, K. Holden, and J. Butler. 2003. *Universal Principles of Design*. Rockport Publishers.
- [16] K. Kuutti. 1996. Activity theory as a potential framework for human-computer interaction research. In B. A. Nardi (Ed.), *Context and Consciousness: Activity Theory and Human-Computer Interaction* (pp. 17-44). MIT Press.
- [17] A. Lucero, A. Desjardins, C. Neustaedter, K. Höök, M. Hassenzahl, and M. E. Cecchinato. 2019. A Sample of One: First-Person Research Methods in HCI. In Companion Publication of the 2019 on Designing Interactive Systems Conference 2019 Companion (DIS '19 Companion). ACM, New York, NY, USA, 385–388. <https://doi-org.ezproxy.its.uu.se/10.1145/3301019.3319996>
- [18] Robert J. Moore, Eric Young Liu, Saurabh Mishra, and Guang-Jie Ren. 2020. Design Systems for Conversational UX. In Proceedings of the 2nd Conference on Conversational User Interfaces (CUI '20). Association for Computing Machinery, New York, NY, USA, Article 45, 1–4. <https://doi-org.ezproxy.its.uu.se/10.1145/3405755.3406150>

- [19] B. A. Nardi (Ed.). 1996. Context and Consciousness: Activity Theory and Human-Computer Interaction. MIT Press.
- [20] J. Nielsen. 1994. Usability Engineering. Morgan Kaufmann.
- [21] Preece, J., Rogers, Y., & Sharp, H. (2015). Interaction Design: Beyond Human-Computer Interaction. Wiley. ISBN: 978-1119020752.
- [22] Emma J. Rose, Craig Macdonald, and Cynthia Putnam. 2023. Design Systems: A scalable model for teaching design systems for UX. In Proceedings of the 5th Annual Symposium on HCI Education (EduCHI '23). Association for Computing Machinery, New York, NY, USA, 5–7. <https://doi-org.ezproxy.its.uu.se/10.1145/3587399.3587403>
- [23] M. Suarez. 2019. Design Systems Handbook. InVision, New York. <https://www.designbetter.co/design-systems-handbook>
- [24] J. Yew, G. Convertino, A. Hamilton, and E. Churchill. 2020. Design Systems: A Community Case Study. In Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20